Ethics and Explainability of the Bank Marketing Model

Course: GAI 601 — Machine Learning | Assignment 4: Ethics & Explainability

Assignment 3 produced three tree-based models for the Portuguese bank term-deposit campaign — a Decision Tree (DT), a Random Forest (RF), and an XGBoost (XGB). Among the eighteen experiments I ran across the three algorithms, the Random Forest experiment RF-4 was the overall best model with F1 = 0.486, slightly ahead of XGB-4 at F1 = 0.477, with the winning Decision Tree experiment DT-5 at F1 = 0.413. This assignment goes back to those three winners — DT-5, RF-4, and XGB-4 — and asks a different question. Performance alone is no longer the goal. The goal is to understand what each model has actually learned, whether it is fair, and whether the bank should deploy it. I use SHAP (SHapley Additive exPlanations) as the explainability tool throughout. SHAP gives mathematically-grounded, per-prediction attributions that can be aggregated to produce global rankings or kept at the customer level for individual decisions — both views are needed and both are presented below.

Because the Decision Tree was the core required algorithm in Assignment 3, I give it the most detailed walkthrough. The Random Forest (RF-4) is identified throughout as the best-performing model and is what the bank would actually deploy. XGBoost (XGB-4) is covered for comparison and to confirm that the three architectures converge on the same fundamental story about which features matter.

1. SHAP Global Explainability

Global SHAP averages the absolute contribution of each feature across many customers — answering the question, "what does the model rely on overall?" I computed SHAP values on a 500-customer stratified sample of the test set (small enough for clean plots, large enough for stable rankings) and produced a summary bar plot and beeswarm for each of the three models. The bar plot ranks features by mean absolute SHAP value (impact magnitude); the beeswarm adds direction — red dots are high feature values, blue dots are low, and dots to the right push the prediction toward "subscribe".

Top 3 features — Decision Tree (DT-5)

Feature 1 — econ_pressure (engineered macro-economic feature, combining consumer price index and consumer confidence). This is by far the dominant feature in DT-5. The beeswarm shows that customers contacted during periods of low econ_pressure (favorable macro conditions) are much more likely to subscribe, while those contacted during periods of high econ_pressure (anxious consumers, inflationary stress) refuse. The model has correctly recognized that term deposits are a "safe haven" product — people park money in CDs when they feel confident about the economy and hoard cash when they do not. This is benign and matches business intuition.

Feature 2 — cons.conf.idx (raw consumer confidence index). A sibling of econ_pressure. The tree extracts additional non-linear signal from the raw confidence index even after the engineered feature is in the model. The bank should consider pausing campaigns during confidence troughs — call-center capacity is wasted when consumer confidence is very low.

Feature 3 — contact_telephone (landline vs cellular). This is where SHAP earned its keep. The beeswarm shows that landline contacts get a strong negative SHAP — the model has learned to de-prioritize landline

customers. As a campaign optimization signal this is useful: cellular calling converts better. But landlines correlate strongly with older, less-affluent, more-rural customers. This is the first major fairness flag, which I unpack in Section 3.

I cross-checked the SHAP ranking against the tree itself. The features that DT-5 splits on most often in the top five levels of the tree are the same features SHAP identifies as globally dominant. SHAP did not invent a story; it recovered the tree's actual logic from the outside. That match is the kind of audit guarantee a regulator wants.

Top 3 features — Random Forest (RF-4, the best model)

The top three features for RF-4 are essentially the same: econ_pressure (#1), contact_telephone (#2), cons.conf.idx (#3). The magnitudes are smoother than DT-5 because the ensemble of 300 trees averages out the single tree's tendency to over-concentrate. The ranking is therefore more reliable: a single tree can be fragile (change a few rows and it picks different splits), but RF-4's SHAP ranking is an average across 300 differently-trained trees. The fact that RF-4 — the model the bank would actually deploy — also ranks contact_telephone in its top three makes the fairness concern in Section 3 concrete rather than hypothetical.

Top 3 features — XGBoost (XGB-4) comparison

XGB-4 produces the same top three: econ_pressure, contact_telephone, cons.conf.idx. The magnitudes are larger because XGBoost outputs log-odds rather than probabilities, but the ordering is identical. The convergence across three architecturally different algorithms — single tree, parallel ensemble, sequential ensemble — is strong evidence that the ranking reflects real signal in the data, not an artefact of any one algorithm.

2. SHAP Local Explanation

Where global SHAP answers "what does the model rely on overall?", local SHAP answers "why this particular customer?" To demonstrate the contrast I picked the test-set customer that the best model (RF-4) most confidently predicts as a subscriber: a 70-year-old retired married man with basic primary education, an existing housing loan, no credit-default history, contacted on his cellphone on a Tuesday in August. Importantly, the previous campaign was already successful for this customer — he subscribed before. RF-4 predicts him to subscribe with probability 0.97, and the true label is indeed yes. This is the kind of correctly-explained correct prediction the bank wants in its audit log.

Local SHAP for DT-5 (deep walkthrough)

The waterfall plot starts at the model's base value (the average predicted probability) and shows each feature pushing the prediction up (toward "yes") or down (toward "no") until it lands at the final probability. For this customer DT-5 shows four large positive pushes: (i) econ_pressure is very low (favorable economy), (ii) poutcome_success = 1 (he subscribed in the prior campaign), (iii) previously_contacted = 1 (he is not a cold call), and (iv) age = 70 (the model gives older customers a small positive push, matching the retiree-buys-CDs pattern). A handful of small dummy variables (e.g.,

month_aug) contribute mild negative drags, but none are large enough to override the positive signal. The final predicted probability is essentially 1.0.

What is striking about this local explanation is that it gives exactly the reasons a human relationship manager would give if asked. The model is not making the right prediction for spurious reasons — it has recovered the same defensible business logic a human would apply.

Local SHAP for RF-4 (best model)

The RF-4 waterfall for the same customer tells essentially the same story but with the contributions smoothed across 300 trees. The largest positive push is poutcome_success = 1, followed closely by econ_pressure being very low. previously_contacted, age, and contact = cellular (i.e., contact_telephone = 0) each contribute additional positive pushes. The local explanation is more stable than DT-5's — a single tree can produce jumpy explanations if a feature value moves it across a split, while the ensemble produces smooth, defensible attributions. For a customer-facing decision the bank has to defend, that stability matters.

Local SHAP for XGB-4

XGB-4 agrees with the other two models. The same set of features (econ_pressure, poutcome_success, previously_contacted, age, contact channel) drives the prediction, in roughly the same order. The XGBoost magnitudes are on the log-odds scale so they appear larger numerically, but the direction and ordering of contributions is consistent with DT-5 and RF-4.

Local vs global — the key contrast

Globally the dominant feature across all three models is econ_pressure. Locally, for this particular customer, the dominant feature is poutcome_success — the fact that he subscribed before. A global feature ranking would have told the bank "focus on the economy." But for this specific 70-year-old retired customer, the more honest one-sentence explanation is, "he subscribed last time, and he's being approached during a favorable economic window — call him." That nuance is invisible in the global plot. It only appears when SHAP is computed at the individual level. This is why local explanations are indispensable for any model that produces customer-facing decisions: the average across customers is not the explanation any one customer is owed.

3. Ethics and Fairness

How SHAP supports ethical AI

SHAP advances three pillars of ethical AI. First, transparency: every Random Forest or XGBoost prediction can be accompanied by a short, mathematically-grounded report saying which features drove that specific prediction. That report is what the EU AI Act and CFPB / ECOA fair-lending regulations increasingly require. Second, bias detection: SHAP's mean absolute attribution gives the true average contribution per feature with direction information, which is the only honest way to ask "does the model rely on a feature I do not want it to rely on?" That is exactly how I detected the contact_telephone issue below. Third, trust: an agent calling a customer can be given a one-paragraph reason for the call, derived from the local SHAP values, which both builds trust with the customer and gives the agent context for the conversation.

Bias risks the SHAP analysis surfaced

The SHAP results highlighted three concrete bias risks in the model. (1) contact_telephone (landline vs cellular) is in the top three globally for all three models, and the data confirms that landline customers are systematically older and lower-income and subscribe at a much lower rate. The model is not lying about the pattern, but by acting on it the bank ends up giving fewer term-deposit offers to exactly the demographic that often most needs a safe savings product. This is the textbook definition of indirect discrimination or disparate impact. (2) age is in the top ten for all three models. Older customers receive a positive push (retirees buy CDs, which is benign), but the same feature also de-prioritizes younger adults from a savings product that would help them build wealth. Age-based segmentation is sometimes acceptable and sometimes illegal; the bank's compliance team needs to know the model uses age strongly, which is what SHAP reveals. (3) default_unknown — the 21% of customers who declined to disclose whether they have ever defaulted on credit — appears in the top features for RF-4 and XGB-4. The model has learned that non-disclosure is itself predictive, which is a silence-as-signal trap: a customer's choice to protect their financial privacy is being used in a downstream decision, and the non-disclosing group may correlate with cultural background, immigration status, or distrust of financial institutions.

Mitigations the bank should adopt before deployment

The three risks above each have a fix. The cleanest fix for contact_telephone is to drop it from the feature set and accept the small F1 drop in exchange for removing the proxy. An alternative is to keep it but apply a fairness post-processing layer that enforces equal recall across age bands. For age, re-derive it into business-relevant brackets and have the marketing team explicitly approve the targeting policy rather than letting the model implicitly construct one. For default_unknown, replace the binary unknown-as-its-own-category with an explicit "disclosure rate" feature so the model can still use the signal but the bank can audit it.

Reflection on Assignments 2 and 3 — what I would do differently

Doing this SHAP audit while the Assignment-3 modelling work is fresh exposes several places where I should have acted differently earlier in the process. In Assignment 2 (cleaning), I treated missing data and the unknown token as data-quality issues. They are also fairness issues — I should have computed who is in the "unknown" group and asked whether the missingness itself correlates with a protected characteristic before deciding to keep "unknown" as a category. I also did not audit demographic representation in the data itself; if the training set over-represents urban professionals, every model I build will quietly inherit that skew. A simple distribution check by age, region, and occupation would have caught it.

In Assignment 3 (modelling), I tracked accuracy, precision, recall, F1, and ROC-AUC for every experiment. None of those measures whether the model treats subgroups equitably. I would add equal-opportunity-difference (recall gap between age groups) and demographic-parity-difference (predicted-positive-rate gap between groups) to the evaluation row, alongside F1. I would also run a small SHAP audit during experimentation rather than only after — adding "top-3 features" to the results log would have surfaced the contact_telephone issue at experiment RF-1 rather than at Assignment 4. Finally, the feature-engineering step that produced econ_pressure was made on predictive grounds; an inclusive process would have included a "who could this hurt?" question for each engineered feature.

Concrete changes for future models: drop or transform proxy features (contact_telephone is the immediate candidate); set fairness thresholds before training rather than after (e.g., recall gap ≤ 10 percentage points across age bands) and reject any model that violates them, regardless of F1; document the model with a Model Card describing intended use, demographic performance, and the SHAP top-features list; re-compute SHAP rankings monthly in production as a drift check; and provide an exception channel so customers the model de-prioritizes can still be reached via a randomized fairness-override fraction of the campaign, both as a fairness mitigation and as ongoing data for the next training round.

Conclusion

All three models — DT-5, RF-4 (the best), and XGB-4 — converge on a defensible global story (the macro economy and contact history drive subscription) and produce sensible per-customer explanations consistent with what a human relationship manager would say. But the same SHAP analysis also reveals that the models rely meaningfully on features that act as proxies for age and socioeconomic status (contact_telephone in particular, with age and default_unknown as secondary concerns). The bank should not deploy the model in its current form without addressing those proxies. SHAP did not solve the fairness problem — it surfaced it. That, in the end, is the value proposition: SHAP makes the model auditable enough that the fairness conversation can actually happen, and concrete enough that the fixes can be specific rather than vague. The deepest takeaway from this assignment is that performance, explainability, and fairness are not three separate evaluation criteria but three views of the same question — what has the model actually learned, and is that something we are willing to deploy.